

VSEC

```
git --version
```

```
git config --global user.name "Snehapawar29"
```

```
git config --global user.name
```

```
git config --global user.email "snehapawar2905@gmail.com"
```

1. Perform the following tasks using git-bash
a. Initialize a Git repository for a data science project
b. Push it to GitHub to enable version control
c. Create a python file name as program1.py and write a python program to print multiplication table of any number
d. Commit the code and push the changes and check the changes on remote repository(GitHub)
e. Create a new branch named GIT-EXAM and create new file named as program2 and write a program to find greatest of two numbers
f. Commit the changes and push them and demonstrate the difference of changes in master and GIT-EXAM branch

✓ a. Initialize a Git repository

Open **Git Bash** and run:

```
mkdir data-science-project
```

```
cd data-science-project
```

```
git init
```

✓ b. Push it to GitHub

1. Go to GitHub and create a new repository (e.g., data-science-project)
2. Connect local repo to GitHub:

```
git remote add origin https://github.com/your-username/data-science-project.git
```

OR

```
git clone https://github.com/your-username/data-science-project.git
```

```
cd data-science-project
```

✓ c. Create program1.py (Multiplication Table)

```
touch program1.py
```

code.

Open file and write:

```
# program1.py
```

```
num = int(input("Enter a number: "))
```

```
for i in range(1, 11):
```

```
    print(num, "x", i, "=", num * i)
```

✓ d. Commit and push changes

```
git add .
```

```
git commit -m "Added program1 for multiplication table"
```

```
git branch -M main
```

```
git push -u origin main
```

👉 Now check your repository on GitHub — file should be visible.

✅ e. Create new branch GIT-EXAM & program2

```
git checkout -b GIT-EXAM
```

```
touch program2.py
```

```
code.
```

Write code:

```
# program2.py
```

```
a = int(input("Enter first number: "))
```

```
b = int(input("Enter second number: "))
```

```
if a > b:
```

```
    print("Greatest number is:", a)
```

```
else:
```

```
    print("Greatest number is:", b)
```

✅ f. Commit, push & show difference

Commit and push:

```
git add .
```

```
git commit -m "Added program2 for greatest of two numbers"
```

```
git push -u origin GIT-EXAM
```

🔍 Show difference between branches:

```
git diff main GIT-EXAM
```

- program2.py exists only in **GIT-EXAM**
- Not present in **main**

🔄 Switch branches to verify:

```
git checkout main    ls    👉 Only program1.py visible
```

```
git checkout GIT-EXAM    ls    👉 Both program1.py and program2.py visible
```

How to make program2.py appear in main? You need to merge branches: (OPTIONAL)

```
git checkout main
```

```
git merge GIT-EXAM
```